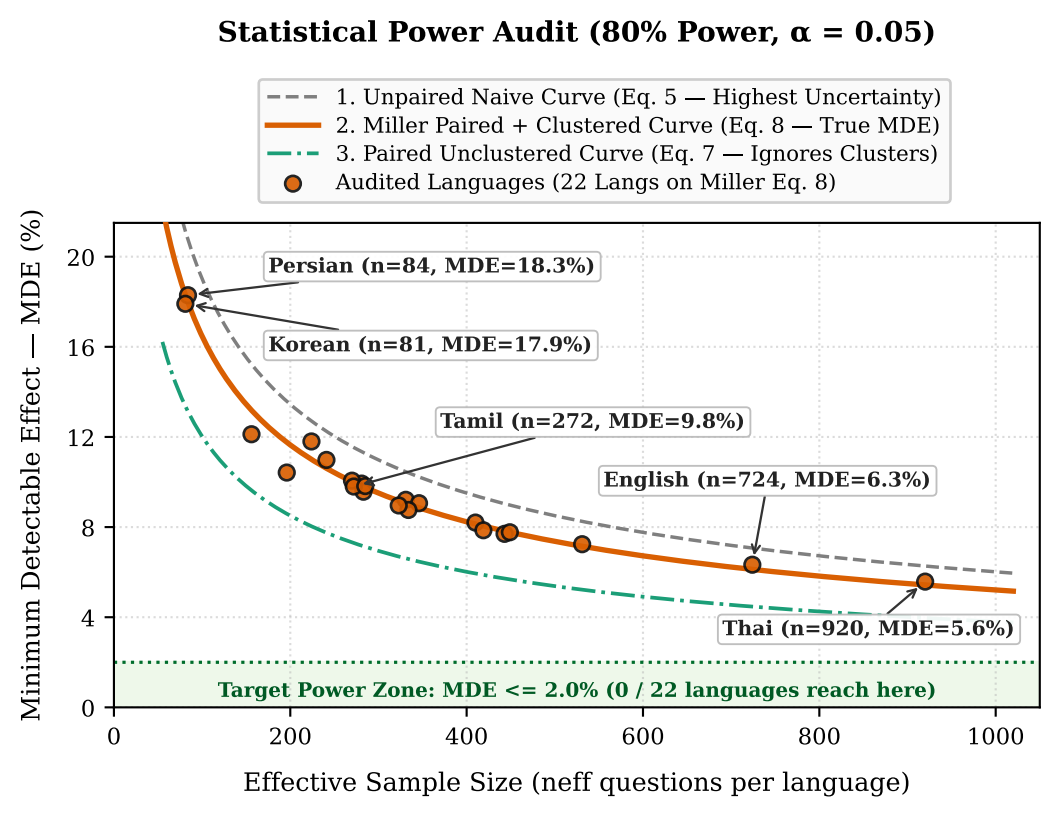
Statistical Power Audit (80% Power, α = 0.05)
1. Unpaired Naive Curve (Eq. 5 — Highest Uncertainty)
2. Miller Paired + Clustered Curve (Eq. 8 — True MDE)
3. Paired Unclustered Curve (Eq. 7 — Ignores Clusters)
Audited Languages (22 Langs on Miller Eq. 8)
Persian (n=84, MDE=18.3%)
Korean (n=81, MDE=17.9%)
Tamil (n=272, MDE=9.8%)
English (n=724, MDE=6.3%)
Thai (n=920, MDE=5.6%)
Target Power Zone: MDE <= 2.0% (0 / 22 languages reach here)
Minimum Detectable Effect — MDE (%)
Effective Sample Size (neff questions per language)
0
4
8
12
16
20
0
200
400
600
800
1000